Neural MS2LDA:
A One-Pass Deep Topic Model for Mass2Motif Discovery

Abstract

Mass2Motif discovery requires a model to explain held-out spectral evidence while retaining a broad inventory of chemically interpretable fragment and neutral-loss patterns. We present a neural probabilistic topic model in which one sparse top-2 routing pass replaces iterative document-level inference. Fixed, train-only token features and learned topic prototypes define both the router and the topic–word decoder. Fragment and neutral-loss channels receive equal probability mass, while balanced assignments, positive-NPMI structure, and prototype separation address topic collapse. The architecture has 233,600 learned parameters and is optimized with alternating router and topic updates over complete spectra. We compare Neural MS2LDA with Tomotopy LDA using the same 1000 topics, seed 42, 6 CPU threads, scaffold split, and leakage-filtered MAG protocol. On validation, the neural model produces 843 optimized motifs and 268 useful high-confidence motifs, versus 607 and 138 for Tomotopy. On test, the corresponding useful-motif counts are 347 and 186. Tomotopy has higher validation mean structure overlap score (SOS), while test mean SOS is tied at four decimals. The result is therefore a breadth–purity trade-off rather than uniform superiority. Recorded core fitting time is lower for the neural model (35.8 versus 76.0 minutes). This is a single-dataset, single-seed research comparison, not a claim that the neural model should replace the production backend.

1 Introduction

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss tokens as words, and recurrent co-occurrence patterns as Mass2Motifs [1]. A useful large-scale model must do more than minimize a predictive loss: it should retain many distinct motifs, assign spectra confidently enough for motif-level analysis, and support chemical annotation [2].

Conventional latent Dirichlet allocation (LDA) estimates a new document's topic mixture by iterative local inference [10]. Amortized neural inference can replace that loop with a learned mapping, but a model with 1,000 topics can collapse: a small subset of topics absorbs most probability mass, several prototypes become near-duplicates, or learned topics ignore observed token co-occurrence. Mass-spectrometry vocabularies introduce another bias. If fragment and neutral-loss channel evidence is summed, the larger channel gains mass simply because it contains more tokens, even when its mean logit is no better.

We address these problems with one shared prototype geometry. It defines a fragment/loss-balanced topic–word distribution and a contextual top-2 router whose aggregated assignments form the document mixture in one pass. The training objective combines full-spectrum likelihood with safeguards against usage, geometric, and co-occurrence collapse. This report states the complete model and evaluation protocol directly.

2 Relation to neural topic models and contribution

The broad ingredients of Neural MS2LDA have established precedents. Neural Variational Document Models introduced an inference network that maps a bag of words to a continuous stochastic document representation [11]. AVITM and ProdLDA subsequently demonstrated one-pass amortized topic inference, discussed component collapse explicitly, and replaced LDA's word-level mixture with a product of experts [12]. The Embedded Topic Model (ETM) parameterizes topic–word natural parameters with inner products between topic and word embeddings [13]. The Neural Sinkhorn Topic Model (NSTM) also learns document-topic weights and cosine topic–word costs in an embedding space, but trains them by minimizing an optimal-transport objective [14]. Contextualized topic models and FASTopic further show how pretrained document representations can enter neural topic discovery [15, 16].

Neural MS2LDA is related to those models, but it is not a VAE, an ETM replica, or an NSTM replica. It contains no sampled document latent variable, posterior family, or KL term. Instead, a spectrum mixture is constructed directly by aggregating sparse, contextual token routes. The same prototypes parameterize those routes and the probabilistic decoder, while Sinkhorn supplies detached balanced training targets rather than the main generative or transport objective. Table 1 makes the distinction explicit.

Model family	Document-topic inference	Topic–word construction	Principal distinction from Neural MS2LDA
LDA / Tomotopy [10, 18]	Iterative inference for each new document	Free categorical topic–word distributions	Neural MS2LDA amortizes inference into one neural pass and shares one geometry between routing and decoding.
NVDM / AVITM / ProdLDA [11, 12]	Document-level neural variational posterior	Bag-of-words decoder; ProdLDA mixes in natural-parameter space	Neural MS2LDA deterministically aggregates token routes and has no sampled posterior, prior-matching KL, or reparameterization step.
ETM [13]	Amortized variational document encoder	Topic and word embedding inner products	Neural MS2LDA shares the geometric decoder idea, but its prototypes also drive sparse token routing with leave-one-out spectrum context.
NSTM [14]	Neural document-topic map	Cosine embedding cost trained through optimal transport	Neural MS2LDA retains mixture likelihood; Sinkhorn only creates detached, balanced router targets.
Contextual / transport embedding models [15, 16]	Pretrained document embedding or direct semantic-relation reconstruction	Neural decoder or learned topic/word embedding transport	Neural MS2LDA uses train-only SGNS token features and a lightweight pooled spectrum context; a pretrained spectrum encoder is future work.
Neural MS2LDA (this work)	Count-weighted aggregation of contextual top-2 token routes	Shared cosine prototypes with fixed 50/50 fragment/loss normalization	Mass-spectrometry-specific coupling of token routing, document gating, channel-balanced decoding, and complementary anti-collapse objectives.

Table 1: Architectural lineage. The claimed contribution is Neural MS2LDA's combined formulation and mass-spectrometry adaptation, not first use of amortization, embedding decoders, sparse routing, or Sinkhorn individually.

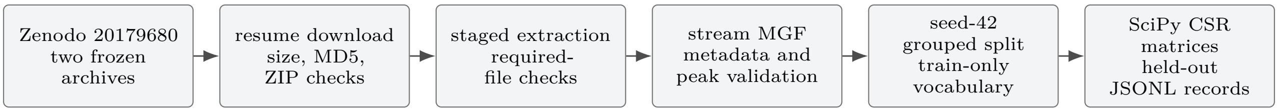

Figure 1: Reproducible MS^{n}Lib ingestion. The MGF supplies spectral documents; the positive-mode Spec2Vec assets branch from the verified extraction only for leakage-filtered MAG annotation.

Within this comparison, no preceding model combines token-level leave-one-out spectrum context, shared router/decoder prototypes, a directly aggregated top-2 document mixture, and fixed fragment/loss channel mass. The methodological contribution claimed here is therefore this mass-spectrometry-specific formulation and its evaluation against Tomotopy. It is not a claim that each building block is new, nor a first-ever priority claim over the entire neural topic-model literature.

3 Dataset and benchmark construction

3.1 MS^{n}Lib source, acquisition, and loading

MS^{n}Lib is an open reference-standard library created from seven compound collections using automated multi-stage fragmentation acquisition and processing. The complete resource contains more than 2.3 million MS^n spectra, including more than 357,000 MS^2 spectra for 30,008 unique structures across both ionization modes [3]. The MS2LDA 2.0 study filtered its positive-mode MS^{n}Lib material to construct a diverse reference-standard benchmark for unsupervised substructure discovery [2]. We use that published benchmark input rather than resampling the full MS^{n}Lib resource.

The exact paper assets are distributed in Zenodo record 20179680 [19]. The acquisition script downloads only `Data.zip` and `model_positive_mode.zip`. It verifies the Zenodo-declared byte size and MD5 of each archive, checks ZIP integrity and paths, extracts through a staging directory, verifies the required files and writes an acquisition manifest. The topic-model input is `Data/Benchmark_MSn_Lib/Corinna_Library_filtered_positive.mgf`; the positive-mode Spec2Vec model, embedding matrix, and SQLite spectral library are loaded only by the downstream MAG evaluation. Thus known molecular structures are used for leakage-safe splitting and chemical scoring, not as labels for fitting either topic model.

At run time, `protocol.json` supplies paths relative to a user-provided data root; absolute-path escape is rejected. The streaming MGF reader treats each `BEGIN IONS`/`END IONS` block as one spectrum, parses peaks rather than trusting `NUM_PEAKS`, and requires a Universal Spectrum Identifier (USI), feature identifier, SMILES, InChIKey, and precursor m/z. RDKit reparses the SMILES and checks that its derived connectivity InChIKey agrees with the deposited identifier. Figure 1 summarizes the hand-off from the public deposit to the sparse matrices consumed by the models.

3.2 Spectrum processing, representation, and split

The source MGF contains 41568 positive-mode spectra. Within each spectrum, intensities are divided by the largest finite peak intensity. Peaks outside $m/z \in (0, 2000]$ or normalized intensity $[0.01, 1]$ are discarded; if more than 1,000 remain, the most intense peaks are kept, and spectra with fewer than five retained peaks are dropped. This deterministic cleaning retains 38888 spectra.

Each physical peak generates a fragment word and, when the precursor-minus-peak mass exceeds 0.01 Da, a neutral-loss word. Both masses are rounded to two decimal places, producing tokens such as `frag@70.04` and `loss@162.05`, as in MS2LDA 2.0 [2]. A peak with normalized intensity I is represented by round$(100I)$ copies of each applicable word. The result is a count-valued bag of spectral words, not a sequence and not a list of structure labels.

Molecules are keyed by the connectivity block of the InChIKey. Cyclic molecules are grouped by non-chiral Bemis–Murcko scaffold [6]; each acyclic connectivity group remains intact. Groups are sorted by descending size and assigned deterministically toward 70/10/20 train, validation, and test targets using seed 42. The resulting matrices contain 27222 training, 3889 validation, and 7777 test spectra, with zero compound or scaffold overlap. The vocabulary is then constructed from training spectra only, requires document frequency at least three, preserves first-seen training order, and contains 21233 fragment/loss tokens. Every split is written in that fixed column order as a SciPy CSR count matrix.

For document-completion evaluation, the loader deterministically divides whole physical peak groups 50/50. A fragment and the neutral loss derived from the same peak therefore cannot fall on opposite sides. The observed half is renormalized without consulting the withheld maximum; out-of-vocabulary held-out words are recorded and excluded from the likelihood denominator. All learned parameters, token features, vocabulary entries, and co-occurrence statistics use the training partition only. Validation and test spectra use that fixed training vocabulary and are evaluated as separate held-out splits.

4 Neural MS2LDA architecture

Neural MS2LDA must connect two directions of reasoning. Given a spectrum, it must infer a small set of topics that explain the observed peaks; given a topic, it must produce an explicit fragment/loss distribution that can be inspected as a Mass2Motif. The subsections follow that forward path. They first define the probabilistic contract, then construct token representations, place tokens and topics in one shared geometry, and finally aggregate context-dependent token routes into a document mixture.

Each piece addresses a separate requirement. Fixed train-only features give the model co-occurrence information without using molecular labels. Shared prototypes keep routing and decoding semantically aligned, while channel balancing prevents the larger fragment or neutral-loss vocabulary from winning by size alone. The contextual sparse router resolves tokens that have different meanings in different spectra and replaces iterative document inference with one neural pass. Together these pieces retain a conventional probabilistic topic–word output while providing a differentiable spectrum–topic map.

4.1 Model notation and overview

Let $d \in \{1, \ldots, D\}$ index spectra, $w \in \{1, \ldots, V\}$ index the shared fragment/loss vocabulary, and $k \in \{1, \ldots, K\}$ index topics. The count $c_{dw} \geq 0$ is the spectral-word multiplicity. The model returns a document–topic distribution $\theta_d \in \Delta^{K-1}$ and a topic–word distribution $\beta_k \in \Delta^{V-1}$, with

$$p(w \mid d) = \sum_{k=1}^{K} \theta_{dk} \beta_{kw}. \tag{1}$$

4.2 Fixed token features

Each token has a fixed feature vector $x_w \in \mathbb{R}^{50}$. Its first 48 coordinates are skip-gram negative-sampling (SGNS) embeddings trained only on training documents [7]. For five epochs, eight count-weighted positive pairs are sampled per document; each positive pair uses five negatives drawn with probability proportional to corpus frequency raised to 0.75. Source embeddings are initialized uniformly on $[-0.5/48, 0.5/48]$ and context embeddings at zero. SparseAdam uses learning rate 0.01 and batches of 8,192 pairs. Source and context embeddings are averaged and normalized after the fifth epoch. Two final one-hot coordinates mark fragment versus neutral loss. The normalized SGNS vector and type indicators are concatenated and normalized again. These are the complete token features used by the model; molecular labels do not enter fitting.

4.3 Shared topic geometry and balanced decoder

This stage gives every spectral word and every topic a position in one learned space. Sharing the space is important because the two directions of the topic model must agree: the router assigns observed evidence to a topic, while the decoder states which evidence that topic is expected to emit. If those two parts used unrelated representations, a spectrum could be routed to a topic whose word distribution described something else. The shared prototypes are therefore the semantic bridge between one-pass inference and interpretable Mass2Motifs.

A learned bias-free matrix $W \in \mathbb{R}^{128\times 50}$ projects the fixed features, and every topic has a learned prototype $q_k \in \mathbb{R}^{128}$:

$$z_w = \mathrm{normalize}(Wx_w), \qquad \widehat{q}_k = \mathrm{normalize}(q_k). \tag{2}$$

The matrix W is initialized orthogonally. The same z_w and $\widehat{q}_k$ are used by both decoder and router. Decoder logits are temperature-scaled cosine similarities,

$$\ell_{kw} = \frac{2\widehat{q}_k^\top z_w}{\tau_\beta}, \qquad \tau_\beta = 0.18. \tag{3}$$

Let V_F and V_L denote the fragment and neutral-loss vocabularies. Each channel receives exactly one half of the topic probability mass, and a softmax within that channel determines token rankings:

$$\beta_{kw} = \frac{1}{2}\frac{\exp(\ell_{kw})}{\sum_{v\in V_{t(w)}}\exp(\ell_{kv})}. \tag{4}$$

Thus $\sum_w \beta_{kw} = 1$, vocabulary size cannot change channel mass, rankings within each channel are unchanged, and the channel correction adds no learned parameter. The fixed split is needed because fragments and neutral losses are two chemically useful views of the same peaks but can have very different vocabulary sizes. A single global softmax could reward the larger channel merely for having more candidate words. Giving each channel half of the mass removes that size effect while leaving the model free to learn rankings within each channel. The resulting β_k is both the probabilistic decoder used in the likelihood and the Mass2Motif spectrum supplied to MAG.

4.4 Contextual top-2 router and document mixture

The decoder answers "what does topic k contain?"; the router answers the reverse question, "which topics explain this spectrum?" A fragment or loss can occur in several chemical contexts, so its identity alone is insufficient. The router combines three levels of evidence: the token embedding, the other words in the same spectrum, and a pooled whole-spectrum summary. This lets the same token support different topics in different spectra without running an iterative inference procedure for each new document.

For spectrum d, define the count-weighted embedding sum $s_d = \sum_w c_{dw} z_w$. For each non-zero token entry, the leave-one-out context is

$$\bar{z}_{d\setminus w} = \frac{s_d - c_{dw} z_w}{\max(\sum_v c_{dv} - c_{dw}, 1)}. \tag{5}$$

A nonlinear context router transforms the concatenated token and context. Let $W_1 \in \mathbb{R}^{256\times 256}$ and $W_2 \in \mathbb{R}^{128\times 256}$ denote its learned affine layers. Then

$$e_{dw} = W_2\,\mathrm{LayerNorm}\big(\mathrm{GELU}\big(W_1[z_w;\bar{z}_{d\setminus w}] + b_1\big)\big) + b_2, \tag{6}$$
$$h_{dw} = \mathrm{normalize}(z_w + e_{dw}), \qquad u_d = \mathrm{normalize}(s_d). \tag{7}$$

The final router layer is initialized near zero, so training begins near the token geometry while retaining a fully nonlinear learned correction. The model has 233,600 learned parameters: 6,400 in the token projection, 99,200 in the context router, and 128,000 in the topic prototypes. Local and document evidence are added before routing:

$$a_{dwk} = h_{dw}^\top \widehat{q}_k + u_d^\top \widehat{q}_k, \qquad r_{dw}^{\mathrm{soft}} = \mathrm{softmax}_k(a_{dwk}/T). \tag{8}$$

Only the two largest entries of r_{dw}^{soft} are retained and renormalized to form r_{dw}^{hard}. Training uses the straight-through quantity $r^{\mathrm{hard}} + r^{\mathrm{soft}} - \mathrm{stopgrad}(r^{\mathrm{soft}})$; inference uses r^{hard} directly.

Count-weighted routed mass is $M_{dk} = \sum_w c_{dw} r_{dwk}$. The same whole-spectrum score supplies a multiplicative document gate,

$$g_{dk} = \mathrm{softmax}_k(u_d^\top \widehat{q}_k/T), \qquad \theta_{dk} = \frac{M_{dk} g_{dk}^\gamma}{\sum_j M_{dj} g_{dj}^\gamma}, \qquad \gamma = 0.75. \tag{9}$$

The gate is detached in Equation 9 during training, so it cannot reduce the loss by sharpening itself; its document score still learns through Equation 8. Multiplication preserves exact zero support from token routing. If a spectrum has no routed mass, θ_d is explicitly uniform. Equations 8–9 require one forward pass and no per-spectrum optimization loop. Conceptually, contextual tokens cast sparse topic votes, count aggregation combines those votes, and the document gate checks whether the voted topics also fit the spectrum as a whole. Because the gate only reweights routed mass, it cannot introduce a topic unsupported by the token-level assignments.

The count-weighted completion loss implements Equation 1 exactly at non-zero target entries:

$$\mathcal{L}_{\text{comp}}(\theta, X) = -\frac{1}{\sum_{dw} c_{dw}} \sum_{dw} c_{dw} \log\left(\sum_{k} \theta_{dk}\beta_{kw}\right). \tag{10}$$

Sparse indexing avoids materializing the dense matrix $\theta\beta$ but does not change its probabilities. Read from left to right, the complete forward path is: fixed token features → learned token positions → context-dependent top-2 routes → one document mixture → a mixture of topic-word distributions. The prototype bank is the shared junction: it receives routing evidence from the left and defines the decoder on the right. Figure 2 shows this flow and separates it from the training-only safeguards.

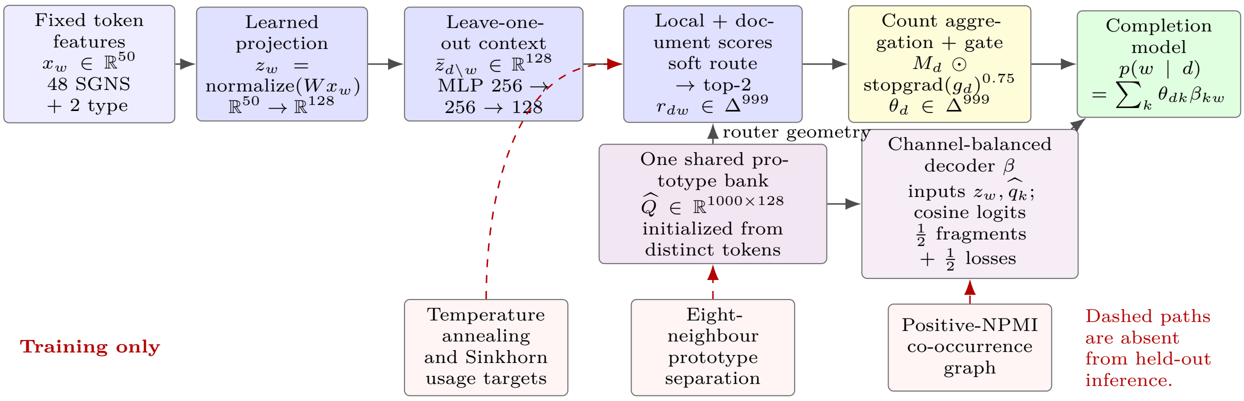

Figure 2: Complete Neural MS2LDA computation graph. Solid arrows form the one-pass forward model used at training and inference. The prototype bank branches to both the contextual router and the probabilistic decoder. Dashed red arrows show training-only collapse safeguards. Dimensions and channel normalization are part of the implemented architecture, not illustrative placeholders.

5 Training objective and anti-collapse design

Defining the forward model is not sufficient when 1000 topics share one learned space. Without coordinated learning, a few topics can absorb most routes, several prototypes can become duplicates, or a decoder can assign high probability to words that rarely co-occur. This section therefore separates two training questions: how the router and decoder take turns learning from the same completion objective, and how the model preserves a broad, coherent topic inventory while they do so.

The first subsection describes the alternating router and topic blocks. This schedule is needed because the projection and prototypes define both sides of the model: changing the routes changes the evidence used to learn topics, while changing the decoder changes what counts as a good route. The second subsection collects safeguards with distinct jobs—initialization anchors topics, temperature annealing moves from exploration to specialization, Sinkhorn resists topic starvation, positive NPMI rewards observed co-occurrence, and prototype separation resists duplication. Completion likelihood remains the predictive anchor throughout. These mechanisms affect fitting only; the trained model still uses the single forward path in Figure 2.

5.1 Alternating full-spectrum optimization

The projection and prototypes serve both routing and decoding, so learning one side changes the target seen by the other. Alternating the updates makes that coordination explicit: the router first learns where spectral evidence should go while the current decoder is fixed, then the topic block learns what each topic should emit while the current routes are fixed. The completion likelihood is the common contract between the blocks. Alternation is a training schedule, not an additional inference stage; a fitted model still uses the single forward path in Figure 2.

Optimization alternates two gradient blocks using one AdamW optimizer. In the *router block*, β is cached and detached. One full-spectrum route is optimized with likelihood, Sinkhorn targets, and prototype separation:

$$\mathcal{L}_{\text{router}} = \mathcal{L}_{\text{comp}} + \omega_S\mathcal{L}_{\text{Sinkhorn}} + 5\mathcal{L}_{\text{sep}}. \tag{11}$$

For a mini-batch with N routed entries, log-space Sinkhorn iteration projects $\exp(a/\varepsilon)$, with $\varepsilon = 0.05$, to detached targets Q whose rows sum to one and whose columns sum to N/K [8, 9]. $\mathcal{L}_{\text{Sinkhorn}}$ is cross-entropy from Q to $\mathrm{softmax}(a/T)$. Sinkhorn is a training target only; it is not part of held-out inference.

In the *topic block*, routes are recomputed without gradients and held fixed. Decoder geometry is then optimized with

$$\mathcal{L}_{\text{topic}} = \mathcal{L}_{\text{comp}} + \mathcal{L}_{\text{NPMI}} + 5\mathcal{L}_{\text{sep}}, \tag{12}$$

using the same full training spectra. A full router pass with batch size 128 is followed by four topic-block updates with batch size 512 per epoch. Each auxiliary loss is attached to the part whose failure it addresses: Sinkhorn encourages the router to use the available topic capacity, positive NPMI makes decoder words cohere according to observed co-occurrence, and prototype separation discourages topics from becoming geometric duplicates. Completion likelihood and separation appear in both blocks, keeping prediction and topic identity coupled throughout training.

5.2 Complementary anti-collapse terms

The safeguards target complementary collapse modes: topic starvation, incoherent word groups, and duplicate prototypes.

1. **Deterministic initialization.** Seed 42 selects 1000 distinct projected training tokens uniformly without replacement, giving every initial prototype a concrete spectral-word anchor.
2. **Annealed sparse routing.** Routing temperature falls linearly from 0.5 to 0.1 over 30 epochs. Early soft assignments permit exploration; later top-2 assignments specialize.
3. **Balanced usage.** The Sinkhorn coefficient is 0.25 for the first ten epochs and then decreases linearly to 0.05 at epoch 40. It resists early winner-take-all collapse without forcing uniform usage at the end of training.
4. **Train-only co-occurrence structure.** Binary training documents define a positive-NPMI graph. Tokens require document frequency at least ten, pair frequency at least three, and mutual membership among at most 16 strongest neighbours. If A is the resulting weighted sparse adjacency, the retained edge weight is

$$A_{wv} = \frac{\log[p(w,v)/(p(w)p(v))]}{-\log p(w,v)}, \qquad A_{wv} > 0, \tag{13}$$

where probabilities are empirical document frequencies. Only mutual top-neighbour edges remain. The training loss is

$$\mathcal{L}_{\text{NPMI}} = -\frac{1}{K}\sum_k \log\left(\sum_{wv} \beta_{kw} A_{wv} \beta_{kv}\right). \tag{14}$$

This favours probability mass on training-supported token pairs.
5. **Prototype separation.** For every prototype, the eight nearest other prototypes are penalized above cosine margin 0.3:

$$\mathcal{L}_{\text{sep}} = \mathrm{mean}_{k,j}\left[\max(0, \widehat{q}_k^{\top}\widehat{q}_j - 0.3)\right]^2. \tag{15}$$

This detects geometric duplication that usage balance cannot detect.
6. **Numerical controls.** PyTorch deterministic algorithms are required under the six-thread allowance. Non-finite losses or gradients stop training immediately, and gradient norm is clipped to 5.0. Training always runs to the predeclared epoch 40.

AdamW uses learning rate 10^{-3}, weight decay 10^{-4}, and seed 42. Table 5 lists the complete settings.

6 Evaluation protocol

6.1 Comparator and timing boundary

Tomotopy 0.13.0 [18] is trained independently on the same training matrix and never acts as a teacher. It uses $K = 1000$, scalar initial $\alpha = 0.6$ with optimization interval 10, $\eta = 0.1$, seed 42, 6 workers, and parallel scheme 3. Training advances in blocks of 50 to at most 5,000 iterations and stops when all of the last ten relative perplexity changes are below 0.005. The recorded fit stops at 750 iterations. Held-out Tomotopy mixtures use 100 iterations, 6 workers, parallel scheme 1, and independent-document inference. The neural model has no corresponding iteration count because its document mapping is amortized into one routing pass.

Reported fitting time is optimization-loop wall time. For the neural model it excludes MGF processing, SGNS feature construction, co-occurrence graph construction, initialization, and evaluation; for Tomotopy it excludes sparse matrix expansion, model construction, and evaluation. This gives like-for-like core fitting time but is not total workflow time.

Runtime is reported descriptively and is not a motif-quality metric. Warm inference is reported separately because it measures resident models and already prepared sparse batches; it is not end-to-end latency.

6.2 Predictive, inventory, and chemical metrics

No single number answers whether an unsupervised topic model is useful. The benchmark therefore separates predictive fit, the size of the annotatable motif inventory, and the chemical consistency of that inventory. Table 2 gives the reading rule for every headline metric.

Metric	Calculation	Interpretation
Completion NLL	Log loss on withheld in-vocabulary token counts after routing the observed half-spectrum.	Lower is better predictive fit.
Optimized motifs	Topics for which MAG optimization retains at least one fragment or loss.	Larger means a broader annotatable inventory.
MAG coverage	Optimized motifs divided by all $K = 1000$ topics.	A percentage form of the optimized count.
SOS-evaluable motifs	Optimized topics having at least one high-confidence associated reference-standard molecule with a valid fingerprint.	Denominator for the chemical-quality summary.
Mean/median SOS	Containment of the MAG consensus fingerprint in associated reference molecules, averaged first per motif and then across evaluable motifs.	Higher means greater chemical consistency.
Useful motifs	Evaluable motifs with SOS ≥ 0.6.	Count above the paper's intermediate-or-high boundary; not a manual annotation claim.
Core fitting time	Wall time inside each optimizer loop.	Lower is faster; reported only as context.

Table 2: What each reported metric measures. Counts and SOS answer different questions and should not be collapsed into one ranking.

Document-completion likelihood. For each held-out spectrum, the model sees one deterministic half of its physical peaks and produces $\widehat{\theta}_d^{\text{obs}}$. The other half is scored without updating model parameters:

$$\text{NLL}_{\text{comp}} = -\frac{\sum_{d,w} c_{dw}^{\text{held}} \log\left(\sum_k \widehat{\theta}_{dk}^{\text{obs}} \beta_{kw}\right)}{\sum_{d,w} c_{dw}^{\text{held}}}. \tag{16}$$

The weighting is by withheld token multiplicity, so an intense peak contributes more than a weak peak. Only tokens in the frozen training vocabulary enter the numerator and denominator. NLL assesses prediction of unseen spectral evidence; it does not directly assess whether a topic has a recognizable substructure.

MAG and SOS, step by step. Chemical scoring follows the MS2LDA 2.0 Mass2Motif Annotation Guidance (MAG) and substructure overlap score (SOS) design [2]. MAG uses Spec2Vec representations to retrieve spectrally related reference compounds [4]; SOS compares their molecular fingerprints rather than their peak lists. The benchmark executes the following fixed sequence:

1. **Remove chemical leakage.** Every Spec2Vec-library row sharing a validation or test connectivity InChIKey is removed before building the FAISS search index. The model can therefore not retrieve the evaluated molecule as its own annotation evidence.
2. **Turn a topic into a query.** The 20 highest-probability fragment or loss tokens in β_k form a Mass2Motif pseudo-spectrum.
3. **Retrieve and refine.** MAG searches up to 1,000 neighbours, keeps at most ten unique molecules, applies the published clustering/refinement procedure at cosine threshold 0.9, and optimizes the motif against the selected cluster. If no fragment or loss survives optimization, the topic is not an optimized motif. Annotation coverage is the fraction of all 1,000 topics that do survive.
4. **Build a motif fingerprint.** Each clustered MAG molecule is converted to a MACCS structural-key fingerprint [5]. A bit is retained in consensus fingerprint A_k when it occurs in at least 80% of the cluster, mirroring the MS2LDA 2.0 motif-fingerprint construction [2].
5. **Find molecules that use the motif.** Full-spectrum topic mixtures are used only after fitting. Molecule i is associated with topic k when $\theta_{ik} \geq 0.5$. Repeated spectra with the same connectivity InChIKey are collapsed so one frequently measured standard cannot dominate the result.
6. **Compute containment and summarize.** With B_i the MACCS fingerprint of an associated molecule,

$$s_{ki} = \frac{|A_k \cap B_i|}{|A_k|}, \qquad \text{SOS}_k = \frac{1}{|C_k|} \sum_{i \in C_k} s_{ki}, \tag{17}$$

where C_k contains unique associated connectivity keys. A motif is SOS-evaluable only if it was optimized and has at least one such molecule with a valid fingerprint.

For a concrete example, suppose MAG's consensus for a motif contains ten MACCS bits and an associated reference compound contains seven of those ten. Its containment score is $7/10 = 0.7$; extra bits in the full molecule are not penalized because SOS asks whether the proposed substructure is contained in the molecule, not whether both fingerprints describe identical molecules. Averaging these scores over the unique associated compounds gives that motif's SOS. The paper-defined bands are high (> 0.8), intermediate ($0.6 \leq \text{SOS} \leq 0.8$), and low ($< 0.6$) [2]. We call the high and intermediate motifs "useful" for counting purposes. SOS $= 1$ means complete containment of the consensus fingerprint; it is not proof of an exact substructure, a fragmentation pathway, or a correct annotation without expert review.

7 Results

Figure 3 shows the principal result as a sequence of nested filters: all optimized topics are candidates for annotation, a subset has a high-confidence reference-standard association and can receive SOS, and a further subset reaches the useful SOS boundary. Figure 4 separates chemical consistency, held-out prediction, and core fitting time so that gains in one quantity are not mistaken for gains in all three.

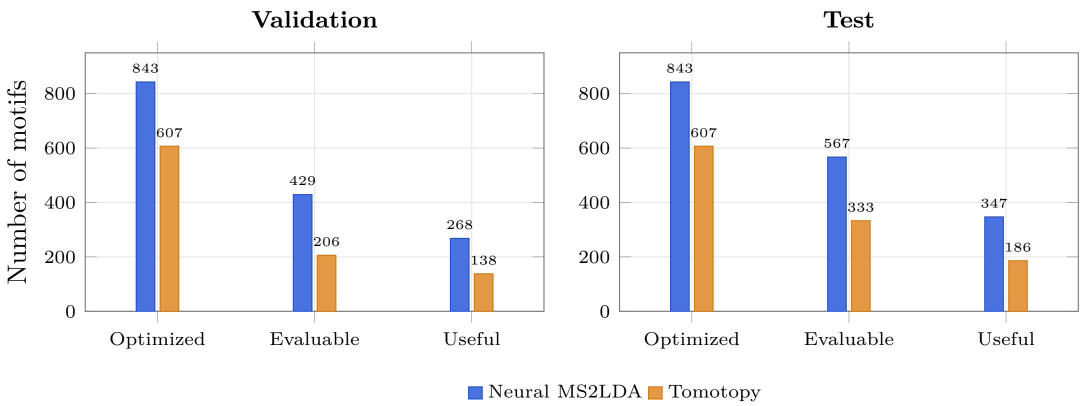

Figure 3: Motif inventory after successive evaluation filters. Neural MS2LDA produces more optimized, SOS-evaluable, and useful motifs than Tomotopy on both held-out splits. Evaluable and useful counts depend on the split-specific compounds; optimized counts depend only on the fitted topic–word distributions and MAG.

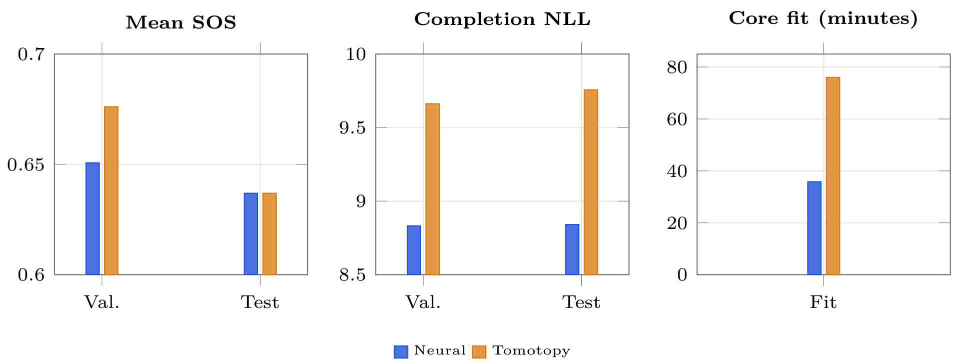

Figure 4: The comparison is a trade-off rather than a single leaderboard. The SOS panel deliberately magnifies the 0.60–0.70 interval; SOS itself lies on $[0, 1]$. NLL is count-weighted document-completion loss. Fitting time excludes preprocessing, feature construction, initialization, and evaluation.

7.1 Validation comparison

Table 3: Primary validation comparison.

Metric	Neural	Tomotopy
Optimized motifs	843	607
MAG annotation coverage	84.3%	60.7%
High-confidence SOS-evaluable motifs	429	206
SOS high (> 0.8)	71	58
SOS intermediate (0.6–0.8)	197	80
SOS low (< 0.6)	161	68
Useful motifs (high-confidence; SOS ≥ 0.6)	268	138
Mean SOS (high-confidence)	0.6507	0.6761
Median SOS (high-confidence)	0.6441	0.6854
Model-fitting time (minutes)	35.8	76.0

Neural MS2LDA produces 843 optimized motifs versus 607 for Tomotopy, a gain of 236 motifs and 23.6 percentage points of MAG coverage. After requiring a high-confidence associated molecule and a valid SOS, the counts are 429 and 206, a difference of 223. The useful counts are 268 and 138, a difference of 130.

The conditional picture is more nuanced. Among evaluable motifs, 62.5% of neural motifs and 67.0% of Tomotopy motifs reach SOS ≥ 0.6. Tomotopy also has higher validation mean SOS (0.6761 versus 0.6507) and median SOS (0.6854 versus 0.6441). The neural model therefore wins on the number of chemically usable candidates, not on validation purity per evaluable candidate.

Predictive evidence points in the other direction: completion NLL is 8.8320 for neural MS2LDA and 9.6622 for Tomotopy, a lower loss by 0.8302 per withheld token. Core fitting takes 35.8 and 76.0 minutes, respectively, so the recorded Tomotopy loop is 2.12-fold longer. This runtime difference is descriptive rather than a measure of motif quality.

7.2 Test comparison

Table 4: Primary test comparison.

Metric	Neural	Tomotopy
Optimized motifs	843	607
MAG annotation coverage	84.3%	60.7%
High-confidence SOS-evaluable motifs	567	333
SOS high (> 0.8)	83	66
SOS intermediate (0.6–0.8)	264	120
SOS low (< 0.6)	220	147
Useful motifs (high-confidence; SOS ≥ 0.6)	347	186
Mean SOS (high-confidence)	0.6369	0.6369
Median SOS (high-confidence)	0.6333	0.6145
Model-fitting time (minutes)	35.8	76.0

On the test split, Neural MS2LDA yields 567 SOS-evaluable motifs versus 333 (234 more), and 347 useful motifs versus 186 (161 more). Annotation coverage remains 84.3% versus 60.7% because the optimized topic spectra and their MAG annotations are shared across the held-out splits.

Unlike validation, test mean SOS is practically identical at four decimals (0.6369 versus 0.6369); the unrounded Tomotopy value is only 3.5×10^{-6} higher. Neural MS2LDA has the higher median (0.6333 versus 0.6145) and the higher useful-within-evaluable share (61.2% versus 55.9%). Test completion NLL is also lower for neural MS2LDA: 8.8412 versus 9.7569, a difference of 0.9158 per token. These are held-out results from the same MS^{n}Lib collection; they are not an independent-dataset replication or a multi-seed uncertainty estimate.

8 Discussion

8.1 How the model components work together

Neural MS2LDA is a deep neural topic model with an explicit probabilistic output. The learned projection and nonlinear leave-one-out context network route spectral evidence to topic prototypes, while the same prototypes decode interpretable fragment/loss distributions. This shared geometry gives the model a single semantic coordinate system: routing determines which topics explain a spectrum, and decoding determines which words make each topic recognizable as a Mass2Motif. Sparse top-2 routing and the document gate then turn many local token decisions into one document-level mixture in a single pass.

The training terms protect different properties of that model. The fixed 50/50 decoder prevents fragment/loss vocabulary size from controlling channel mass. Sinkhorn balancing discourages the router from starving most of the 1,000 topics; positive-NPMI regularization rewards words that genuinely co-occur in training spectra; and prototype separation discourages multiple topics from representing the same region of the learned space. None of these terms replaces the completion likelihood. Together they make the large topic inventory usable while the likelihood keeps the model accountable for predicting spectral evidence.

8.2 Breadth, purity, and practical value

The central empirical finding is breadth together with comparable test-average chemical consistency. Neural MS2LDA exposes many more topics to MAG, and that advantage survives both the high-confidence association filter and the SOS ≥ 0.6 filter. This matters operationally because annotation work begins with a finite candidate inventory: even a high mean SOS is of limited use if many learned topics cannot be optimized or associated with reference molecules. Conversely, more candidates also create more review burden and more opportunities for spurious motifs. Optimized and useful counts should therefore be read together with the SOS distribution, not in isolation.

The validation result shows this tension directly. Tomotopy's evaluable motifs are chemically purer on average, whereas neural MS2LDA supplies substantially more useful motifs in absolute terms. On test, mean SOS is effectively tied and the neural median is higher, so the larger neural inventory is not accompanied by an obvious average-quality penalty there. The lower completion NLL adds a different kind of support: given half a spectrum, the neural model assigns more probability to its withheld fragment/loss evidence. Neither result implies that every additional topic is a distinct biochemical substructure. It says that the model provides a larger set of MAG-supported hypotheses while predicting held-out spectral words better under this protocol.

8.3 What SOS can and cannot establish

SOS is an intentionally asymmetric containment score. It rewards a MAG-derived consensus feature when that feature appears in a molecule associated with the topic, while allowing the complete molecule to contain many other features. This matches the substructure question, but it also means SOS is not a whole-molecule similarity, an atom mapping, or direct fragment-to-atom evidence. Its value depends on the Spec2Vec reference library, the clustering result, the 80% consensus cutoff, the $\theta \geq 0.5$ association threshold, and the MACCS key vocabulary. The low/intermediate/high bands are useful standardized summaries from the MS2LDA 2.0 study [2], not natural laws.

The leakage filter strengthens the benchmark by preventing exact held-out connectivity matches from entering MAG retrieval, and compound balancing stops replicate-rich standards from dominating SOS. Nevertheless, both models are evaluated against the same positive-mode library-derived chemistry. Performance may differ in noisy biological samples, other instruments, negative ionization, or chemical regions poorly represented by the Spec2Vec library. Manual expert inspection remains necessary before assigning a chemical name to a motif.

8.4 Deep-model extensibility

The neural formulation is valuable beyond the present speed comparison because its context pathway is differentiable and modular. A pretrained spectrum-level encoder such as DreaMS [17] could be inserted through a learned adapter that augments or replaces pooled context u_d, following the general logic of contextual neural topic models [15]. The prototype decoder can remain unchanged, preserving a direct topic–fragment/loss distribution even if the context representation becomes richer. A frozen encoder experiment should precede fine-tuning so that gains can be attributed to representation transfer rather than unconstrained capacity.

Other plausible extensions are a lightweight permutation-invariant set encoder, sparse chemical supervision where labels are genuinely reliable, and GPU-native sparse batches. They are hypotheses, not part of the reported model. The present architecture is deliberately a clean integration point: it is deep enough to accept learned spectral representations without converting Mass2Motifs into an opaque clustering output.

8.5 Scope and limitations

This is a single-dataset, single-seed comparison. Fixing seed 42 makes the two implementations reproducible, but the experiment does not quantify run-to-run uncertainty or establish generalization across libraries. Validation and test are disjoint held-out partitions, yet both originate from the same MS^{n}Lib collection. Replication should therefore use new datasets and multiple seeds.

Timing claims have a similarly narrow boundary. The neural core fit is shorter and its warm resident inference is much faster (Appendix C), but neither includes data acquisition, MGF parsing, feature construction, model loading, MAG retrieval, or cold-start overhead. End-to-end deployment speed will depend on those surrounding stages and hardware. Finally, Tomotopy remains the established production backend; this report demonstrates a promising neural alternative and integration substrate, not a production replacement decision.

9 Conclusion

Neural MS2LDA performs document inference in one contextual neural pass while retaining explicit probabilistic topics. Fixed train-only token features and a nonlinear top-2 router produce document mixtures; shared prototypes define balanced fragment/loss distributions. The differentiable context pathway can accept future learned spectrum embeddings without making Mass2Motifs opaque.

Under the fixed seed-42, $K = 1000$, 6-thread MS^{n}Lib protocol, neural MS2LDA produces a substantially larger MAG-optimizable and useful motif inventory than Tomotopy. It also gives lower document-completion NLL and shorter recorded core fitting time. The chemical quality comparison is deliberately qualified: Tomotopy has higher validation mean and median SOS, while test mean SOS is tied at four decimals and neural MS2LDA has the higher test median. The evidence therefore supports a breadth–purity trade-off with strong predictive performance, not blanket superiority on every metric.

Neural MS2LDA couples shared prototypes, channel-balanced decoding, contextual routing, and anti-collapse objectives for mass spectrometry. One library and seed cannot establish robustness. Multi-dataset replication and a frozen DreaMS adapter study come next. Until then, this remains a research architecture and integration substrate.

A Exact hyperparameters

Table 5: Values generated from `protocol.json`.

Quantity	Value
Seed; topics; CPU threads	42; 1000; 6
Token features	48 SGNS + 2 type = 50 dimensions
SGNS	SparseAdam; 5 epochs; 8 pairs/document; 5 negatives; power 0.75; batch 8192; lr 0.01
Token projection; context router	128; hidden 256; Linear–GELU–LayerNorm–Linear
Decoder	temperature 0.18; separate channel softmax; fixed 50/50 fragment/loss mass
Router	top-2; additive document evidence; gate exponent 0.75
Training spectra	one complete-spectrum path per training pass
AdamW	lr 0.001; weight decay 0.0001; gradient clip 5.0; deterministic PyTorch algorithms
Batches and epochs	router 128; topic 512; 4 topic updates/epoch; 40 epochs
Loss weights	NPMI 1.0; separation 5.0
Routing temperature	0.5 to 0.1 over 30 epochs
Sinkhorn	epsilon 0.05; 5 iterations; weight 0.25 to 0.05
NPMI graph	min DF 10; min pair 3; 16 neighbours; positive NPMI
Prototype separation	8 neighbours; margin 0.3
Tomotopy	alpha 0.6; eta 0.1; parallel 3; step 50; max 5000; stop 10 changes < 0.005; inference 100
MAG/SOS	top 20 motif tokens; search 1000; 10 unique molecules; membership ≥ 0.5; cluster cosine 0.9; fingerprint consensus 0.8

B Equation–implementation correspondence

Table 6: Direct map from the mathematical description to executable code.

Report component	Implementation
Token features x_w	`data.py: build_token_features`
Projected tokens z_w	`model.py: projected_tokens`
Decoder β	`model.py: topic_word_distribution`
Nonlinear leave-one-out context	`model.py: _route_embeddings`
Top-2 routing	`model.py: route`
Gated θ_d	`model.py: aggregate_theta`
Likelihood and regularizers	`objectives.py`
Alternating optimization	`training.py; optimization.py`
Held-out inference	`evaluation.py`
MAG and SOS	`mag.py; chemical.py`

C Secondary diagnostics and practical inference

On test, 344 topics exceed uniform mean corpus usage and the median entropy-derived effective topic count is 6.43 per spectrum. These quantities are collapse checks, not headline chemical outcomes. Completion NLL per token is 8.8320 versus 9.6622 on validation and 8.8412 versus 9.7569 on test (neural versus Tomotopy).

In warm in-memory inference on resident models and batches of 128 spectra, the neural router processes 9093.9 spectra/s versus 30.0 spectra/s for Tomotopy, a 302.6-fold ratio. Both measurements use 6 CPU threads; the neural model uses one routing pass and Tomotopy uses 100 inference iterations. This result is relevant to repeated in-memory batches or a resident service, not end-to-end application latency.

D Executable reproduction

The repository contains one current protocol and one current model artifact. The latter consists only of the model state, vocabulary, and the two inference temperatures. From the repository root:

```
conda env create -f environment-neural-ms2lda.yml
conda env create -f environment-msnlib-mag.yml
conda run -n ms2lda-neural python \
  scripts/download_msnlib_validation_assets.py \
  --data-root /path/to/MSnLib-assets
conda run -n ms2lda-neural python -m benchmarks.neural_ms2lda run \
  --data-root /path/to/MSnLib-assets --run /path/to/run
python scripts/generate_neural_ms2lda_report.py
```

The acquisition command validates the public source data before training. Validation evaluation and MAG scoring finish before test evaluation is invoked. The comparator is trained from the same training matrix, so no external Tomotopy result is required.

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